Code No: 124CN **R15**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year II Semester Examinations, April/May - 2023
COMPUTER ORGANIZATION
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART - A

(25 Marks)

1.a)	Define effective address.	[2]
b)	Discuss about status bit conditions.	[3]
c)	Define Cycle stealing.	[2]
d)	Write the Flowchart for programmed I/O.	[3]
e)	Define locality of reference.	[2]
f)	Describe replacement algorithms for set associate cache.	[3]
g)	Define idle state in 8086.	[2]
h)	Describe extra segment register and its purpose.	[3]
i)	Explain Bit manipulation instructions.	[2]
j)	Write instructions to Load 26 (decimal) into register CX.	[3]

PART-B

(50 Marks)

2.a) What is an interrupt? What are different types of interrupts? Explain them.
b) Describe general register organization with block diagram. [5+5]

OR

3.a) Describe register transfer for the fetch phase with circuit.
b) Explain conditional branch instructions. [5+5]

4.a) Draw a timing diagram for handshaking.
b) Explain DMA controller with block diagram. [5+5]

OR

5. Describe Intel 8089 IOP. [10]

6.a) How many 128×8 RAM chips are needed to provide a memory capacity of 2048 bytes.
b) What additional logic is required to give a no match result for a word in an associative memory when all key bits are zero?
c) Explain associate memory page table. [3+4+3]

OR

7. Explain different types of mapping procedures in cache memory. [10]

8.a) Explain special functions of general registers.
b) Describe flag registers in 8086 CPU. [5+5]

OR

9. Specify the addressing modes in 8086 with example. [10]

10.a) Write an ALP to find factorial of number for 8086.
b) Write an ALP to reverse the given string for 8086. [5+5]

OR

11.a) Write assembly language program to list alphanumeric characters by illustrating JUMP instruction?
b) Write instructions to evaluate the expressions:
a = b + c –d
z = x + y + w – v +u [5+5]

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